TestBoard_ADC124 - automatic PCB diagnostics

15 October 2026 | Connected PZT strips, untouched | LPSPI only | Separate temporary diagnostic firmware

Switching history, recovery and acquisition controls

762 captures across 256 microprograms; three repetitions except three 120-second noise records. 7,919,928 acquired cycles, 7,919,811 received, 117 accounted whole-cycle USB discards. Diagnostic timing is measured separately from production sampling performance.

Hardware	Array A	Array B
External MUX	TMUX1108	TMUX1308A
Bias resistance MUX1 / 2 / 3 / 4	470k / 1M / 470k / 1M	1M / 470k / 1M / 470k
Buffer / ADC	OPA365 / ADC124S101	OPA365 / ADC124S101
External address 6	MUX1-3 tied to VMID; MUX4 sensor	MUX1-3 tied to VMID; MUX4 sensor

Both MUX banks use the 3.3 V TPS7A2033PDBVR LDO. REF2033 provides ADC reference 3.3 V and VMID 1.65 V. There is 100 nF near the reference VMID pad, and 150 pF to ground plus a nearby 100 ohm isolation resistor at each MUX VMID input. Exact electrical capacitor/resistor ordering has not been established from a schematic. Nominal 1 count = 0.805664 mV.

Raw datasets, frozen firmware/settings, complete per-conversion labels, channel statistics and capture accounting accompany this PDF. Baseline uniformity, temporal SD and closeness to nominal VMID are different measurements. All SPI frequencies are requested settings; SCLK was not electrically measured. Logged burst markers include driver overhead.

Measured comparisons and practical implications

Array/MUX	Full 0-6, off: center / spread	Only 0-5, off: center / spread	Held addr2, single ADC input
A/1	2,033.00 / 34.00	2,044.50 / 8.00	2,048.00
A/2	2,014.50 / 36.00	2,045.50 / 7.00	2,048.00
A/3	2,031.50 / 42.00	2,045.50 / 5.00	2,048.00
A/4	2,045.00 / 8.00	2,045.00 / 8.00	2,048.00
B/1	1,923.00 / 43.00	1,921.50 / 14.00	2,048.00
B/2	1,990.50 / 13.00	1,990.50 / 16.00	2,048.00
B/3	1,926.00 / 51.00	1,927.50 / 19.00	2,048.00
B/4	1,994.50 / 3.00	1,995.00 / 9.00	2,048.00

5 us / 16 MHz; all values in counts. Centers/spreads compare the same six physical addresses. Removing address 6 removes the periodic VMID connection only on MUX1-3; MUX4 had a seventh sensor there. This is direct evidence about reset history, with a remaining sweep-period/duty change. The separate equal-slot boundary controls address that timing confound.

If held-address baselines return near midscale while scanning baselines do not, a large static leakage/reference offset is insufficient to explain the scan deficit. Switching history, charge redistribution and read timing remain contributors. The sensor is also continuously connected during the held condition, so its charging duty changes.

Extending the powered EN-disabled interval from 1 ms to 1 s changes Array A first-reconnection codes by -1.0 to +0.0 counts, versus -480.0 to -363.0 counts on B. This is a strong array-dependent retention/reconnection effect; it does not separately identify sensor, MUX or PCB leakage.

In sensor recovery tests, the eighth conversion differs from the first by 2.0 to 9.0 counts. The reference-only burst changes its mean by at most 2.162 count with no external or ADC-input switching. Reading a bank second raises its first mean code by 2.034 to 2.296 counts. A shared conversion-related analog effect therefore contributes; sensor settling alone cannot explain all burst recovery. Reference/supply ratio disturbance is a candidate, not a measured rail-voltage diagnosis.

Adding discarded ADC conversions during an otherwise equal VMID connection changes median MUX baseline centers by at most 1.0 count with 50 us reference slots and 1.0 count with 20 us slots. These are controlled diagnostic schedules; no optimized production speed gain was measured. Swapping which ADC bank is read first changes centers by up to 5.0 counts in the matched-slot tests.

The long unreset scan contains a valid-frame burst and baseline step. Maximum route peak-to-peak ranges are 128 counts (103.1 mV) without reset, 72 counts (58.0 mV) with VMID reset, and 2 counts held. The reset record still contains an isolated spike, despite lower sustained variance. Sequential recordings without vibration monitoring cannot establish the source or prove a mode suppresses all events. Removing address 6 is a diagnostic intervention: it also removes the seventh MUX4 sensor, so it is not a full-50-channel production fix.

Controls and interpretation limits

No quiet-window or spike exclusion: all steady-state raw ADC samples retained.

Timing markers bracket ADC bursts; they are not measurements of analog aperture or GPIO edges.

Diagnostic frame encoding/timestamps add overhead; measured rates are diagnostic rates.

Input order and one-conversion lookahead were validated in a tagged digital model; quiet sensors alone do not validate physical channel identity.

Address 6 is VMID on MUX1-3, sensor on MUX4; software VMID-off still contains reference visits when address 6 is present.

Matched-boundary programs fix seven 100 us slots and a requested 710 us cycle. Boundary phase changes are intentional; sensor-5 exposure differs in hold control.

Recovery uses one ADC input on one bank, fixed sensor/reference slots, and retains eight conversions or waits seven nominal word periods before one conversion.

All leakage gaps use powered MUX isolation; these are reconnection responses, not direct measurements of sensor voltage while disconnected.

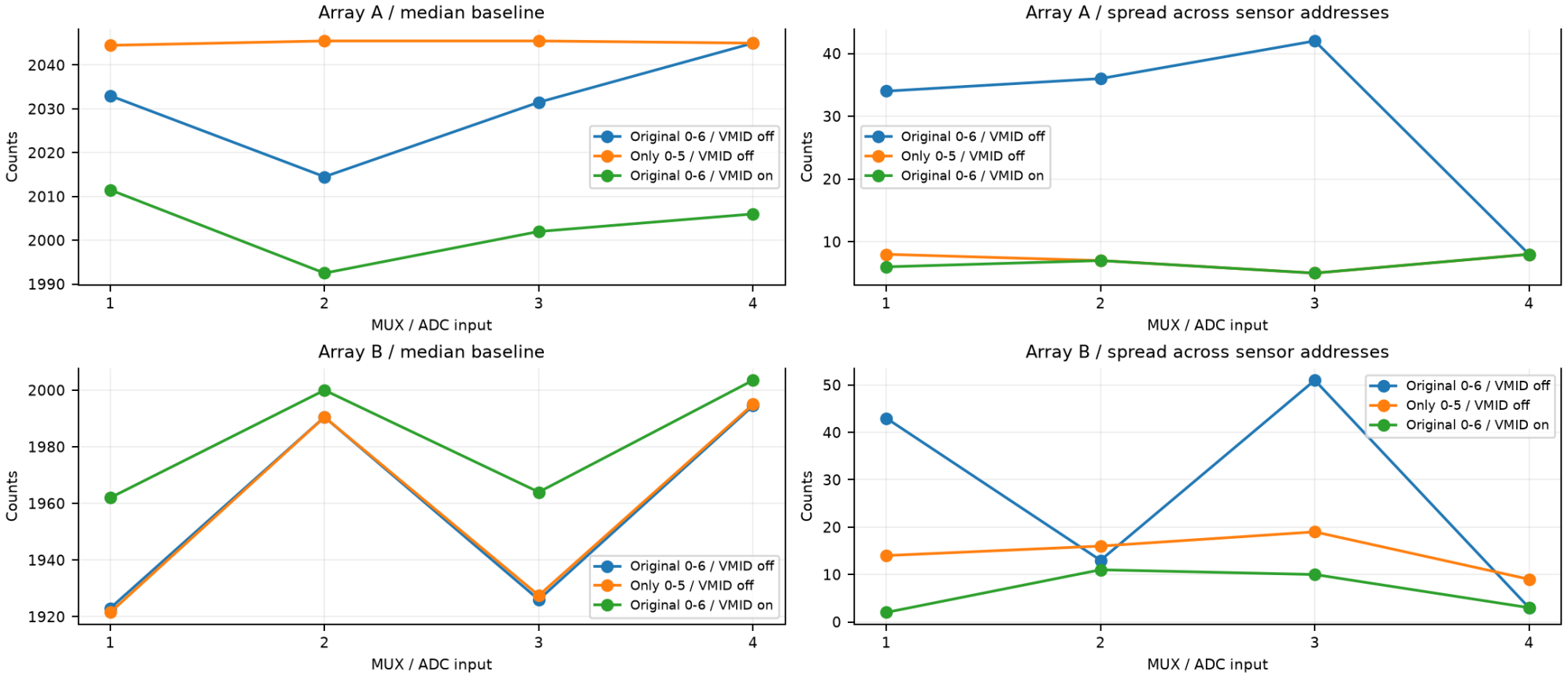
A fixed-address acquisition removes external switching but changes sensor connection duty. It is an empirical static reference, not proof of accuracy under multiplexed operation. Reversing scan order changes both physical predecessors and digital address transitions. A sequence comparison alone cannot isolate digital feedthrough. MUX type remains confounded with array, ADC, buffer and layout.

All initial samples are saved. Baseline/noise statistics exclude the configured warm-up only; leakage tests use no warm-up. Baseline = median of samples, then median of repetition medians. Repeated-setting SD and peak-to-peak are medians of per-record raw statistics; worst-repetition SD and peak-to-peak are also retained in channels.csv. No spikes are removed from the raw data. Sub-count SD is quantization-dependent and is not a direct measurement of intrinsic analog noise.

A separate one-second smoke capture is preserved but excluded from matrix totals. One incomplete long-isolation attempt was archived after a host timeout; the reader was corrected, production restored, and the capture repeated. Only complete, fully accounted captures enter the tables.

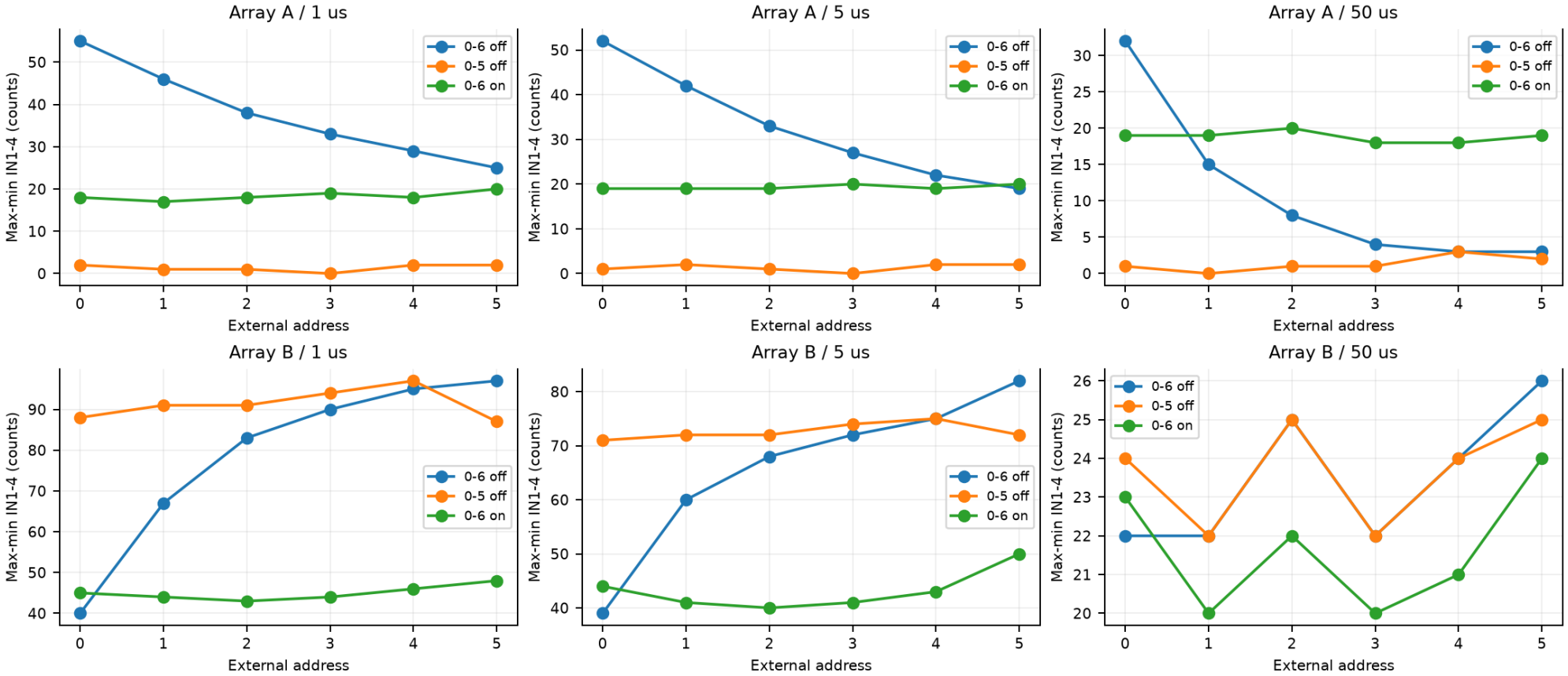
Family	Programs	Captures
boundary_fast	9	27
boundary_matched	15	45
fast_order	3	9
held	15	45
held_extra	1	3
known_vmid	11	33
leakage	32	96
long_noise	3	3
order	10	30
readorder	15	45
recovery	80	240
recovery_bracket	40	120
refburst	7	21
reset_control	9	27
reset_short	6	18

Fast-sequence reproduction and removal of address 6



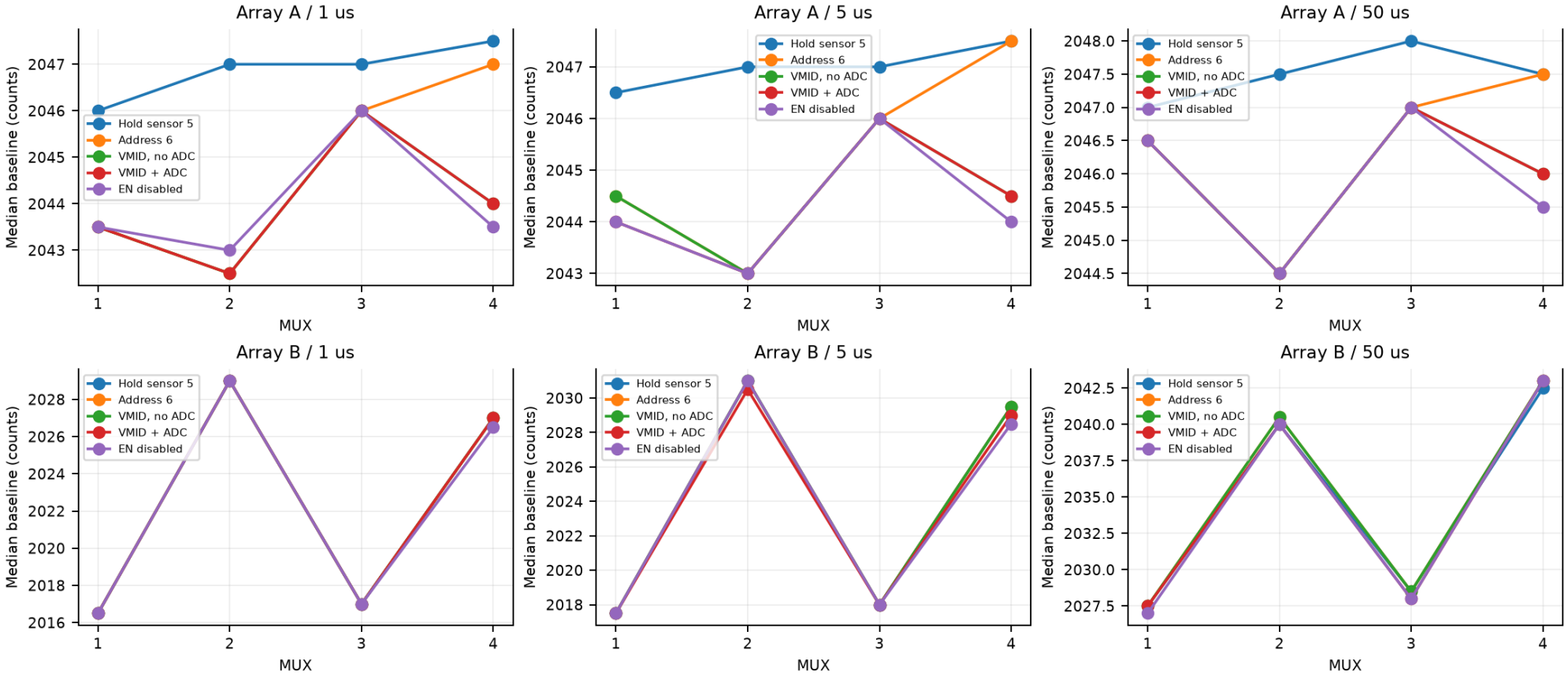
5 us settling, requested 16 MHz. Centers/spreads use addresses 0-5 in every condition. Removing address 6 removes the hidden VMID visit on MUX1-3 as well as MUX4's seventh sensor; sweep duration and exposure also change.

Spread between ADC inputs at each external address



Every point compares four independently estimated baselines at the same address; this is not temporal noise.

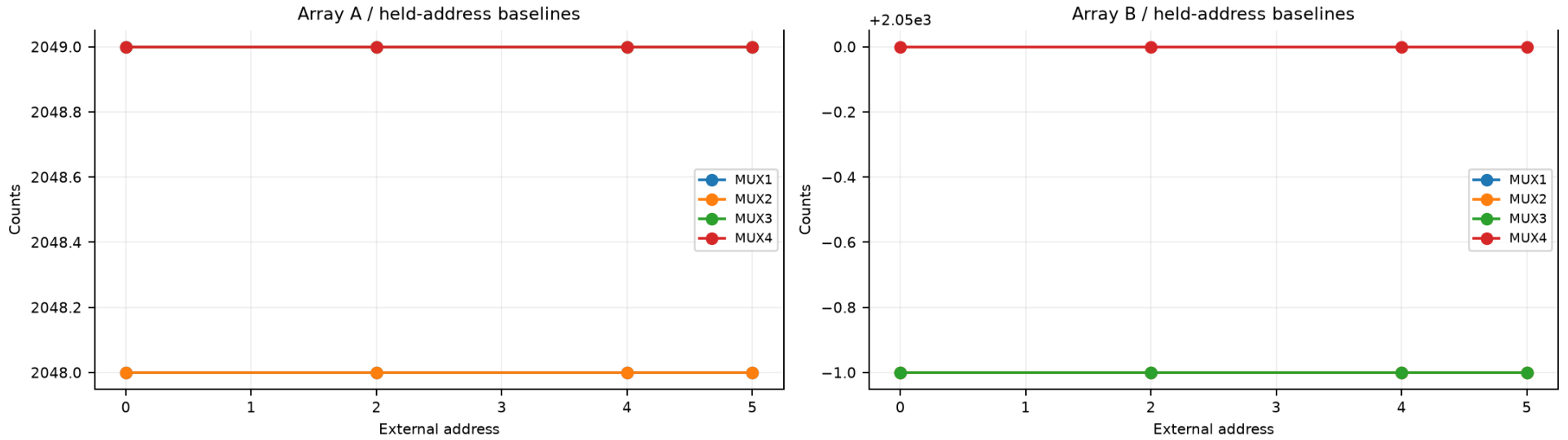
Boundary controls with equal phase slots



Six sensor slots plus one boundary slot, each 100 us; requested period 710 us. The boundary state is intentionally changed. Hold-sensor-5 control exposes that sensor for longer. ADC-on versus ADC-off VMID variants share the same boundary-slot duration, but conversions occur early within that slot.

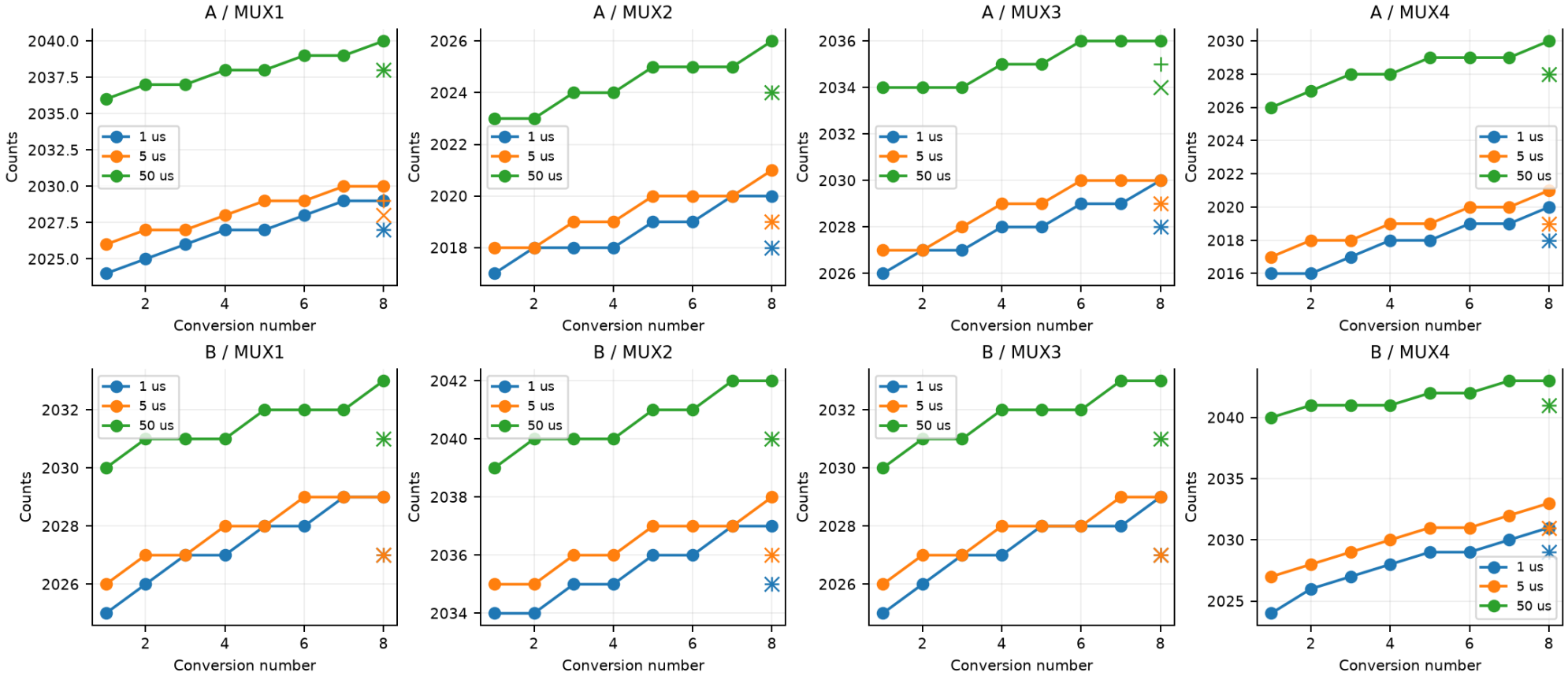
Fixed-address acquisition: external switching removed

Array	MUX	Bias resistor	Scan 0-6 baseline / SD	Held addr2, four ADC inputs	Held addr2, one ADC input
A	1	470k	2,030.00 / 0.027	2,048.00 / 0.000	2,048.00 / 0.000
A	2	1M	2,011.00 / 0.000	2,048.00 / 0.096	2,048.00 / 0.000
A	3	470k	2,029.00 / 0.184	2,049.00 / 0.152	2,048.00 / 0.000
A	4	1M	2,044.00 / 0.000	2,049.00 / 0.000	2,048.00 / 0.000
B	1	1M	1,925.00 / 0.414	2,049.00 / 0.169	2,048.00 / 0.029
B	2	470k	1,989.00 / 0.103	2,049.00 / 0.041	2,048.00 / 0.000
B	3	1M	1,928.00 / 0.048	2,049.00 / 0.000	2,048.00 / 0.053
B	4	470k	1,993.00 / 0.007	2,050.00 / 0.000	2,048.00 / 0.038



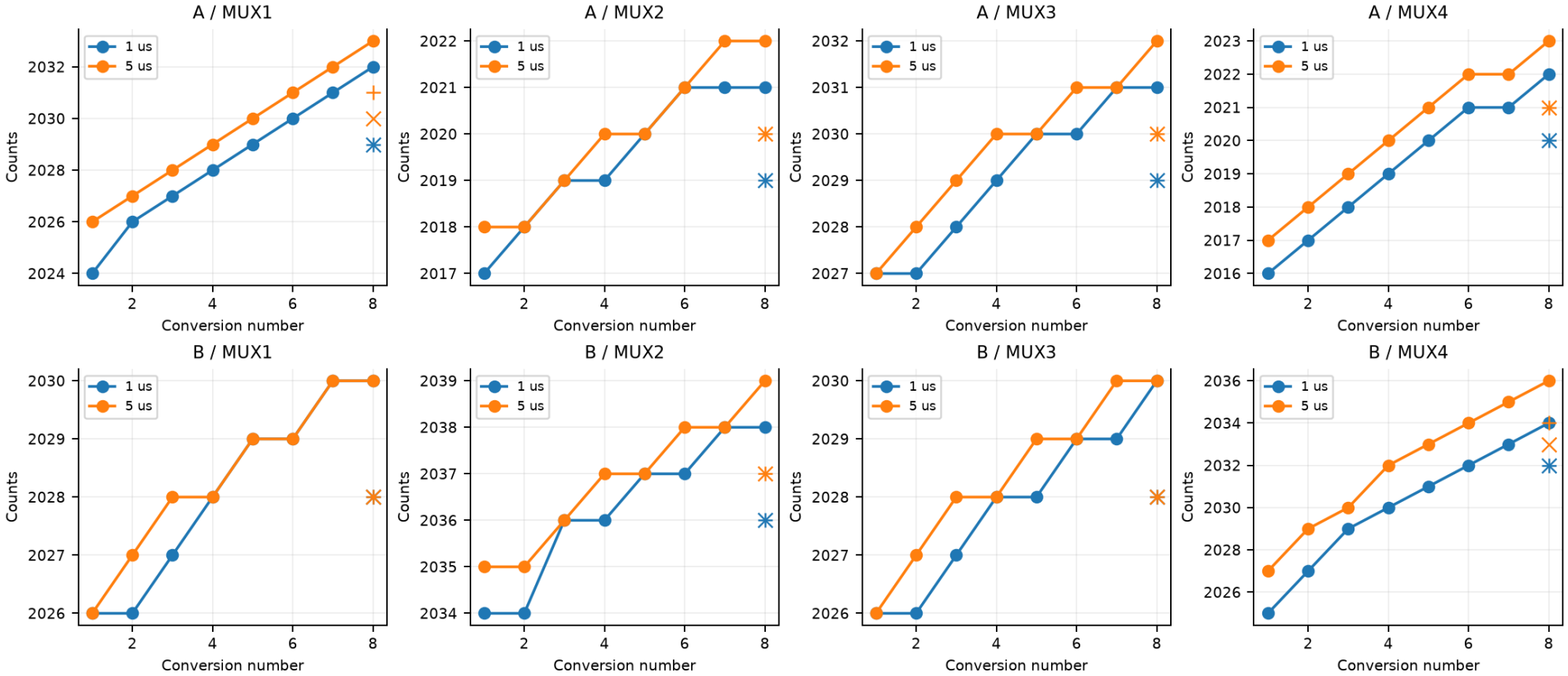
Held programs request 710 us periods. All physical MUXes in a participating bank remain at the chosen address. The four-input versus one-input comparison changes ADC input cycling, while external address remains fixed.

First eight conversions after VMID - 16 MHz



One bank and one ADC input at a time. Reference phase: 100 us wait in a 120 us slot. Sensor phase: fixed 100 us slot, no selector prime inside acquisition. All eight consecutive responses retained. Cross at conversion 8 = single response after waiting seven extra nominal word periods without conversions; plus = one additional us. Markers bracket bursts; exact sample apertures and actual SCLK edges were not probed.

First eight conversions after VMID - 8 MHz



One bank and one ADC input at a time. Reference phase: 100 us wait in a 120 us slot. Sensor phase: fixed 100 us slot, no selector prime inside acquisition. All eight consecutive responses retained. Cross at conversion 8 = single response after waiting seven extra nominal word periods without conversions; plus = one additional us. Markers bracket bursts; exact sample apertures and actual SCLK edges were not probed.

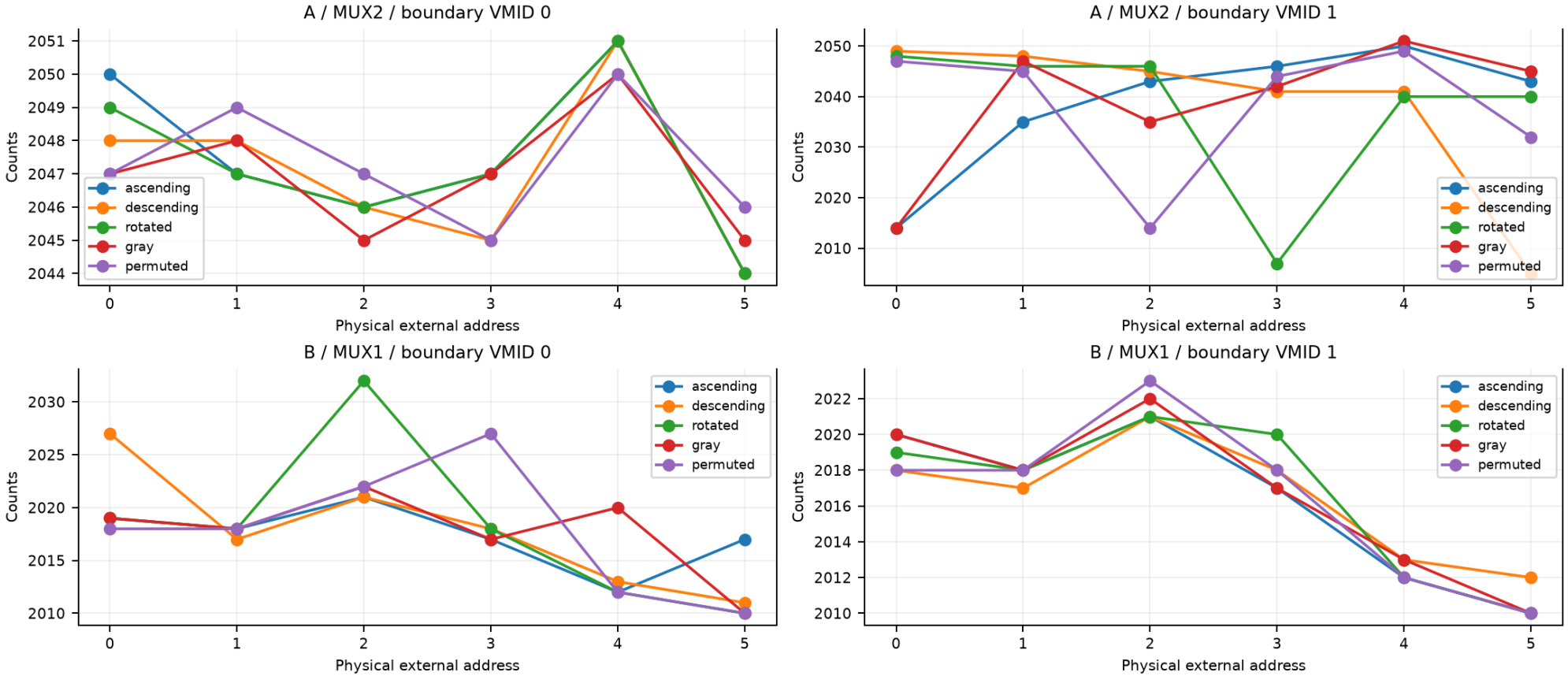
Recovery comparison: conversion activity versus elapsed waiting

Array/MUX	Delay us	MHz	First code	Eighth code	Wait-only code	Eighth-first	Eighth-wait	Burst begins after switch, us
A/1	1	16	2,024.00	2,029.00	2,027.00	5.00	2.00	1.08
A/1	5	16	2,026.00	2,030.00	2,028.00	4.00	2.00	5.08
A/1	50	16	2,036.00	2,040.00	2,038.00	4.00	2.00	50.08
A/1	1	8	2,024.00	2,032.00	2,029.00	8.00	3.00	1.08
A/1	5	8	2,026.00	2,033.00	2,030.00	7.00	3.00	5.08
A/2	1	16	2,017.00	2,020.00	2,018.00	3.00	2.00	1.08
A/2	5	16	2,018.00	2,021.00	2,019.00	3.00	2.00	5.08
A/2	50	16	2,023.00	2,026.00	2,024.00	3.00	2.00	50.08
A/2	1	8	2,017.00	2,021.00	2,019.00	4.00	2.00	1.08
A/2	5	8	2,018.00	2,022.00	2,020.00	4.00	2.00	5.08
A/3	1	16	2,026.00	2,030.00	2,028.00	4.00	2.00	1.08
A/3	5	16	2,027.00	2,030.00	2,029.00	3.00	1.00	5.08
A/3	50	16	2,034.00	2,036.00	2,034.00	2.00	2.00	50.08
A/3	1	8	2,027.00	2,031.00	2,029.00	4.00	2.00	1.08
A/3	5	8	2,027.00	2,032.00	2,030.00	5.00	2.00	5.08
A/4	1	16	2,016.00	2,020.00	2,018.00	4.00	2.00	1.08
A/4	5	16	2,017.00	2,021.00	2,019.00	4.00	2.00	5.08
A/4	50	16	2,026.00	2,030.00	2,028.00	4.00	2.00	50.08
A/4	1	8	2,016.00	2,022.00	2,020.00	6.00	2.00	1.08
A/4	5	8	2,017.00	2,023.00	2,021.00	6.00	2.00	5.08
B/1	1	16	2,025.00	2,029.00	2,027.00	4.00	2.00	1.10
B/1	5	16	2,026.00	2,029.00	2,027.00	3.00	2.00	5.10
B/1	50	16	2,030.00	2,033.00	2,031.00	3.00	2.00	50.10
B/1	1	8	2,026.00	2,030.00	2,028.00	4.00	2.00	1.10
B/1	5	8	2,026.00	2,030.00	2,028.00	4.00	2.00	5.10
B/2	1	16	2,034.00	2,037.00	2,035.00	3.00	2.00	1.10
B/2	5	16	2,035.00	2,038.00	2,036.00	3.00	2.00	5.10
B/2	50	16	2,039.00	2,042.00	2,040.00	3.00	2.00	50.10
B/2	1	8	2,034.00	2,038.00	2,036.00	4.00	2.00	1.10
B/2	5	8	2,035.00	2,039.00	2,037.00	4.00	2.00	5.10
B/3	1	16	2,025.00	2,029.00	2,027.00	4.00	2.00	1.10

Array/MUX	Delay us	MHz	First code	Eighth code	Wait-only code	Eighth-first	Eighth-wait	Burst begins after switch, us
B/3	5	16	2,026.00	2,029.00	2,027.00	3.00	2.00	5.10
B/3	50	16	2,030.00	2,033.00	2,031.00	3.00	2.00	50.10
B/3	1	8	2,026.00	2,030.00	2,028.00	4.00	2.00	1.10
B/3	5	8	2,026.00	2,030.00	2,028.00	4.00	2.00	5.10
B/4	1	16	2,024.00	2,031.00	2,029.00	7.00	2.00	1.10
B/4	5	16	2,027.00	2,033.00	2,031.00	6.00	2.00	5.10
B/4	50	16	2,040.00	2,043.00	2,041.00	3.00	2.00	50.10
B/4	1	8	2,025.00	2,034.00	2,032.00	9.00	2.00	1.10
B/4	5	8	2,027.00	2,036.00	2,033.00	9.00	3.00	5.10

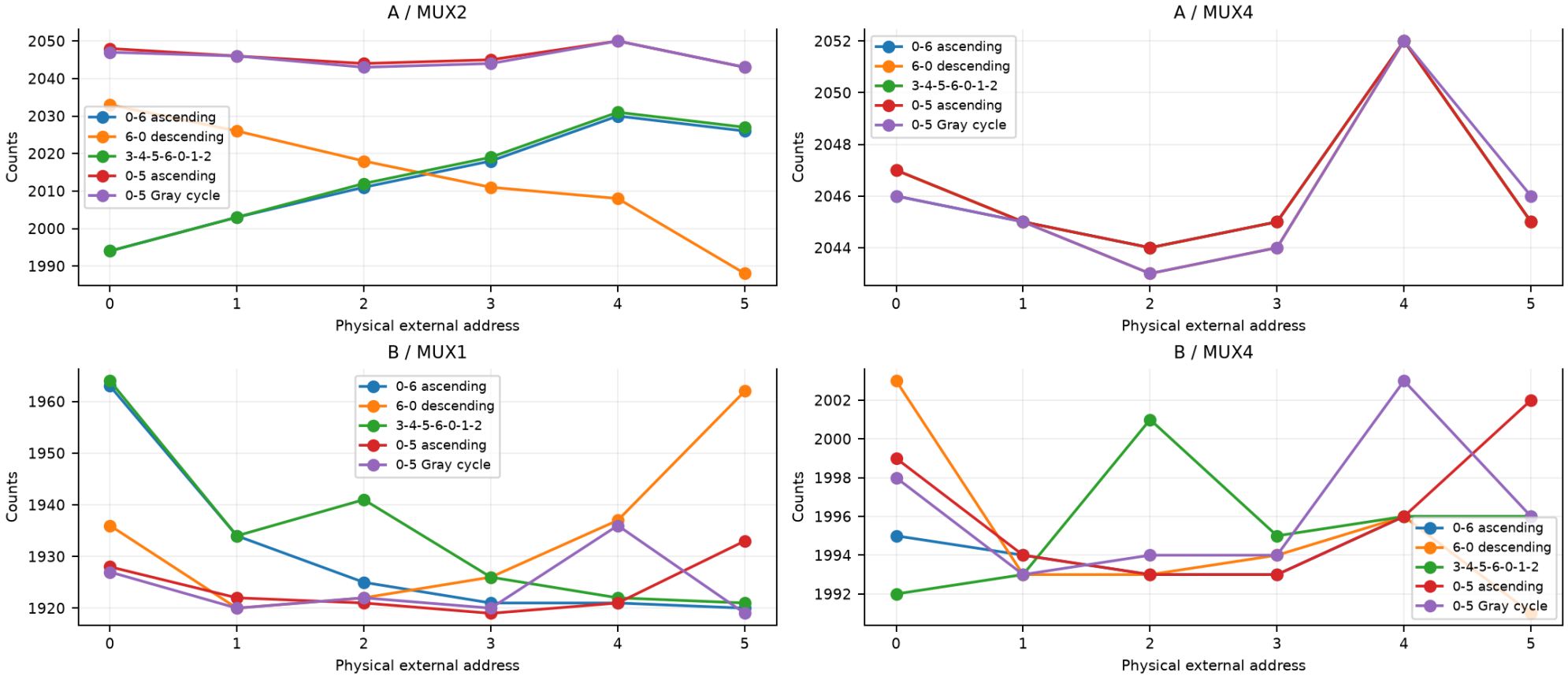
Counts shown are repetition medians. Differences below a count are sensitive to code thresholds. A different wait-only endpoint is a candidate conversion-activity effect, but nominal word spacing is not an independently measured aperture.

Address order: does the pattern follow physical address?



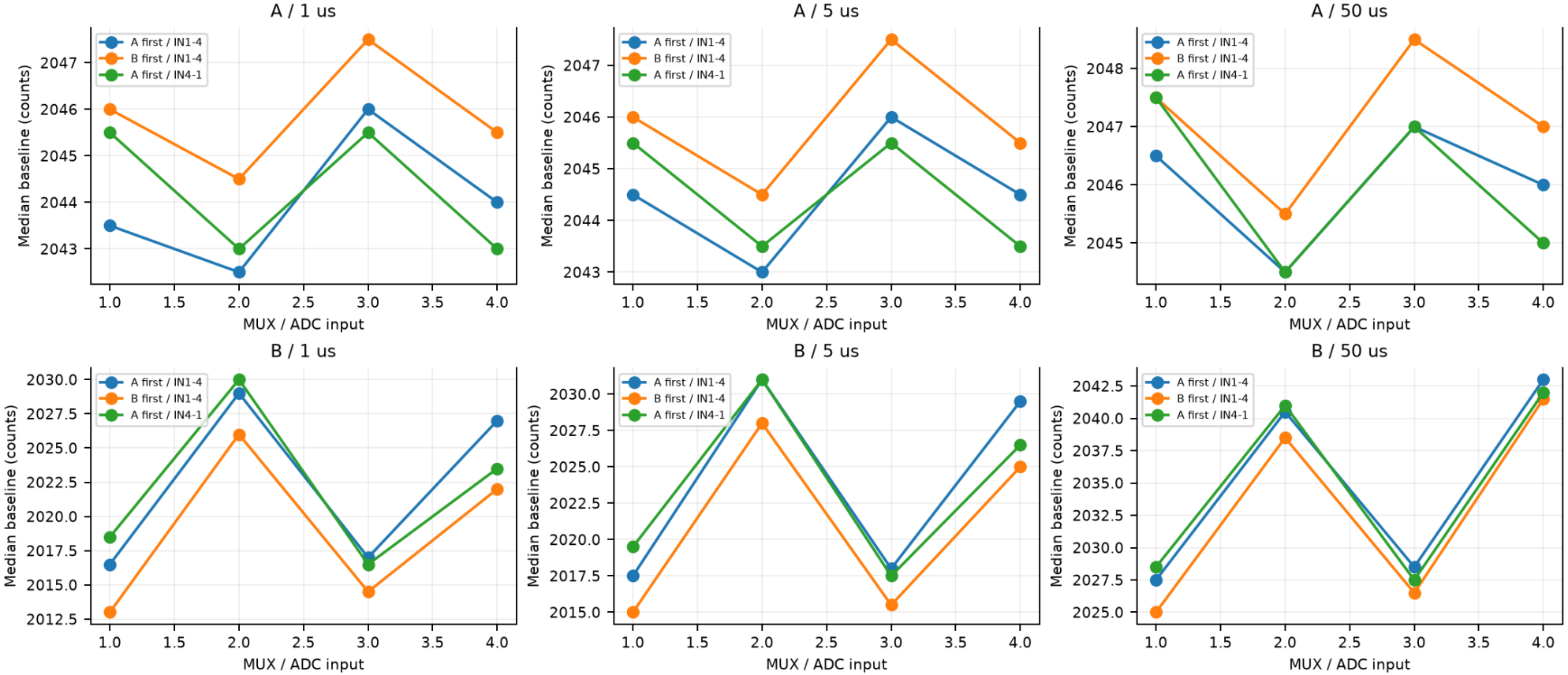
All orders use the same six sensor addresses, 100 us sensor slots and one 100 us boundary slot, period 710 us. Without boundary VMID, the final sensor remains selected during that slot. Gray cycle is 0-2-3-1-5-4-0 (one bit changes each transition). Changing order changes both electrical predecessors and GPIO transitions; it does not isolate their individual contributions.

Fast address order and inter-cycle pause placement



Requested 16 MHz, 5 us, no configured phase slots or cycle throttle. Full descending retains the hidden VMID visit but changes its next sensor from address 0 to 5. Full rotation preserves the cyclic analog predecessor order while moving the USB encoding/enqueue gap from after address 6 to after address 2. Gray versus ascending uses only addresses 0-5. Timing/transition counts may differ; these are diagnostic rates and comparisons, not a production optimization benchmark.

Bank and ADC-input read order



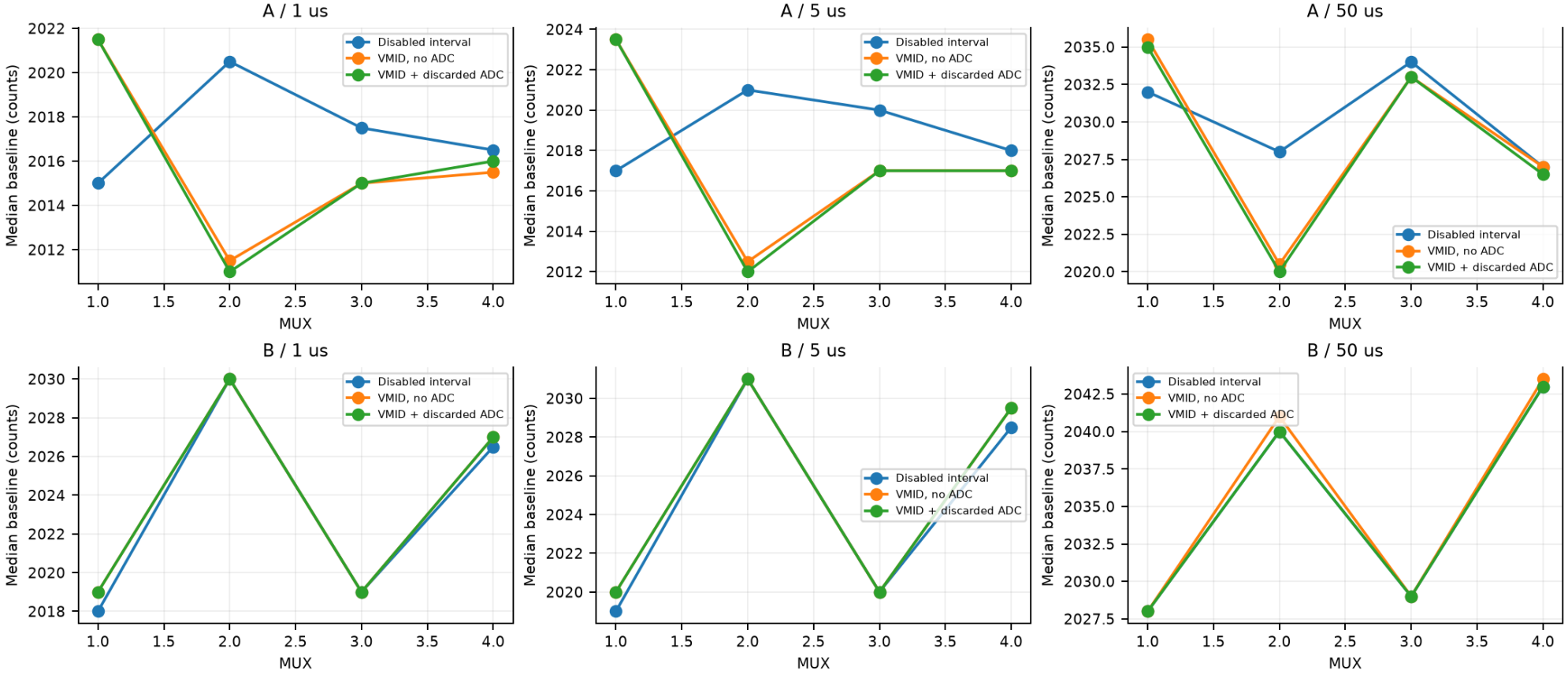
Fixed 100 us slots, six sensor addresses and one boundary VMID visit without ADC activity; period 710 us. Per-bank burst-start timestamps are in channels.csv. Input reversal changes position within the four-word burst as well as ADC predecessor.

Single-bank versus dual-bank activity

Array	MUX	Delay us	Both, A first	Both, B first	Only this bank	Single - matched-first dual
A	1	1	2,043.50	2,046.00	2,044.00	0.50
A	1	5	2,044.50	2,046.00	2,044.50	0.00
A	1	50	2,046.50	2,047.50	2,046.50	0.00
A	2	1	2,042.50	2,044.50	2,043.00	0.50
A	2	5	2,043.00	2,044.50	2,043.50	0.50
A	2	50	2,044.50	2,045.50	2,044.50	0.00
A	3	1	2,046.00	2,047.50	2,046.00	0.00
A	3	5	2,046.00	2,047.50	2,046.50	0.50
A	3	50	2,047.00	2,048.50	2,047.50	0.50
A	4	1	2,044.00	2,045.50	2,044.50	0.50
A	4	5	2,044.50	2,045.50	2,044.50	0.00
A	4	50	2,046.00	2,047.00	2,046.00	0.00
B	1	1	2,016.50	2,013.00	2,013.00	0.00
B	1	5	2,017.50	2,015.00	2,015.00	0.00
B	1	50	2,027.50	2,025.00	2,025.50	0.50
B	2	1	2,029.00	2,026.00	2,026.00	0.00
B	2	5	2,031.00	2,028.00	2,028.00	0.00
B	2	50	2,040.50	2,038.50	2,039.00	0.50
B	3	1	2,017.00	2,014.50	2,014.50	0.00
B	3	5	2,018.00	2,015.50	2,015.50	0.00
B	3	50	2,028.50	2,026.50	2,026.50	0.00
B	4	1	2,027.00	2,022.00	2,022.00	0.00
B	4	5	2,029.50	2,025.00	2,025.00	0.00
B	4	50	2,043.00	2,041.50	2,041.50	0.00

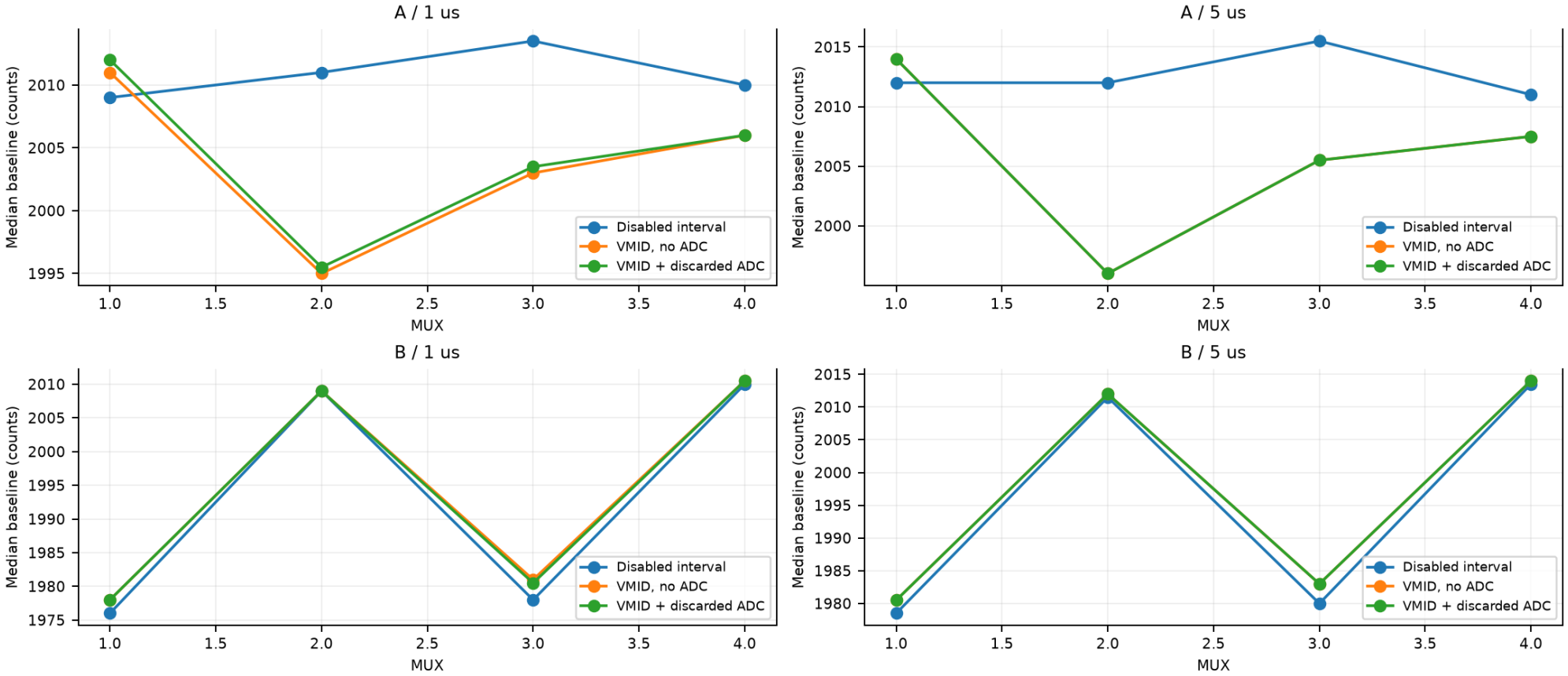
Comparing single-bank to dual-bank with the same bank read first reduces the extra-dwell confound. The inactive bank is EN-disabled. Coordinated versus single-bank GPIO preparation still differs slightly.

VMID clamp versus discarded conversions, with controlled sensor dwell



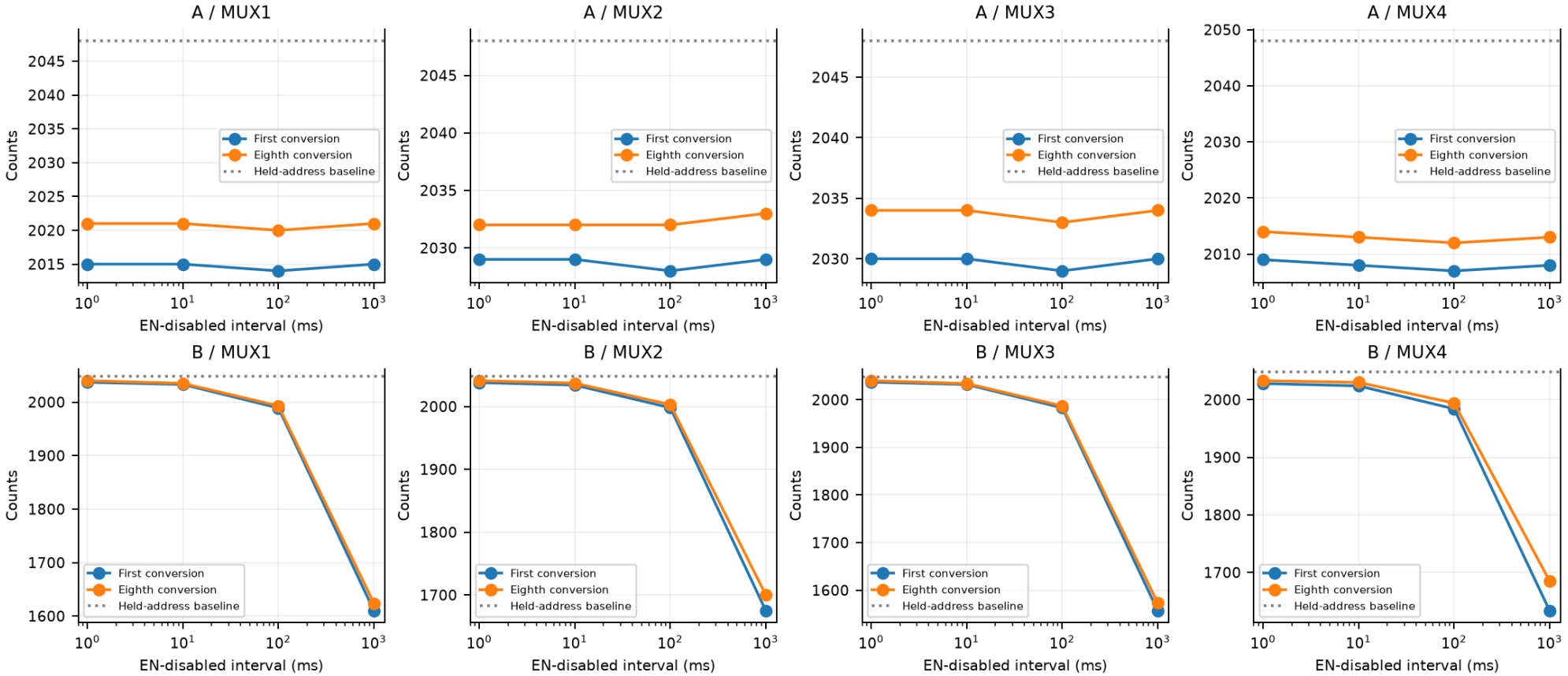
Sensor connection is held for a 100 us slot and disabled at its end. The following interval is 50 us, either disabled throughout or connected to VMID until its end. VMID-with-ADC executes four discarded conversions per bank within the same slot. Cycle period is 920 us. Even the disabled common output remains connected to its bias resistor.

Shorter clamp control: the same comparison at faster cycling



Sensor connection 25 us, then disabled at slot end. Reference/disabled interval 20 us; cycle period 300 us. Both VMID variants keep the reference connected for the same 20 us. This is a separate controlled schedule, not an optimized production driver. It does not demonstrate that all production VMID conversions can be removed without retesting at the production dwell/cadence.

Controlled isolation and reconnection



Address 2 is connected for a 100.1 ms preconditioning slot, then the bank is EN-disabled for the specified interval. Reconnect at address 2, wait 1 us, and retain eight conversions of one ADC input. Six cycles or more per repetition. The common-node bias continues during isolation; initial sensor voltage is not directly measured. These curves do not uniquely determine PZT leakage or capacitance.

Retention response and unchanged-address EN control

Array/MUX	First code, 1 ms off	First code, 1 s off	Change counts	Change mV	Eighth code, 1 s off	1 s first-code range across reps
A/1	2,015.00	2,015.00	0.00	0.00	2,021.00	3.00
A/2	2,029.00	2,029.00	0.00	0.00	2,033.00	0.00
A/3	2,030.00	2,030.00	0.00	0.00	2,034.00	0.00
A/4	2,009.00	2,008.00	-1.00	-0.81	2,013.00	2.00
B/1	2,037.00	1,610.00	-427.00	-344.02	1,624.00	8.50
B/2	2,038.00	1,675.00	-363.00	-292.46	1,700.00	2.50
B/3	2,037.00	1,557.00	-480.00	-386.72	1,573.00	6.00
B/4	2,028.00	1,633.50	-394.50	-317.83	1,685.00	41.50

The address cache stays at 2 throughout the disconnected interval. Reconnection changes EN but does not rewrite address pins. An early-reading deficit in this experiment therefore does not require an address-bit transition. EN switching, charge redistribution on reconnection and ADC history remain coupled; this does not measure EN injection charge separately.

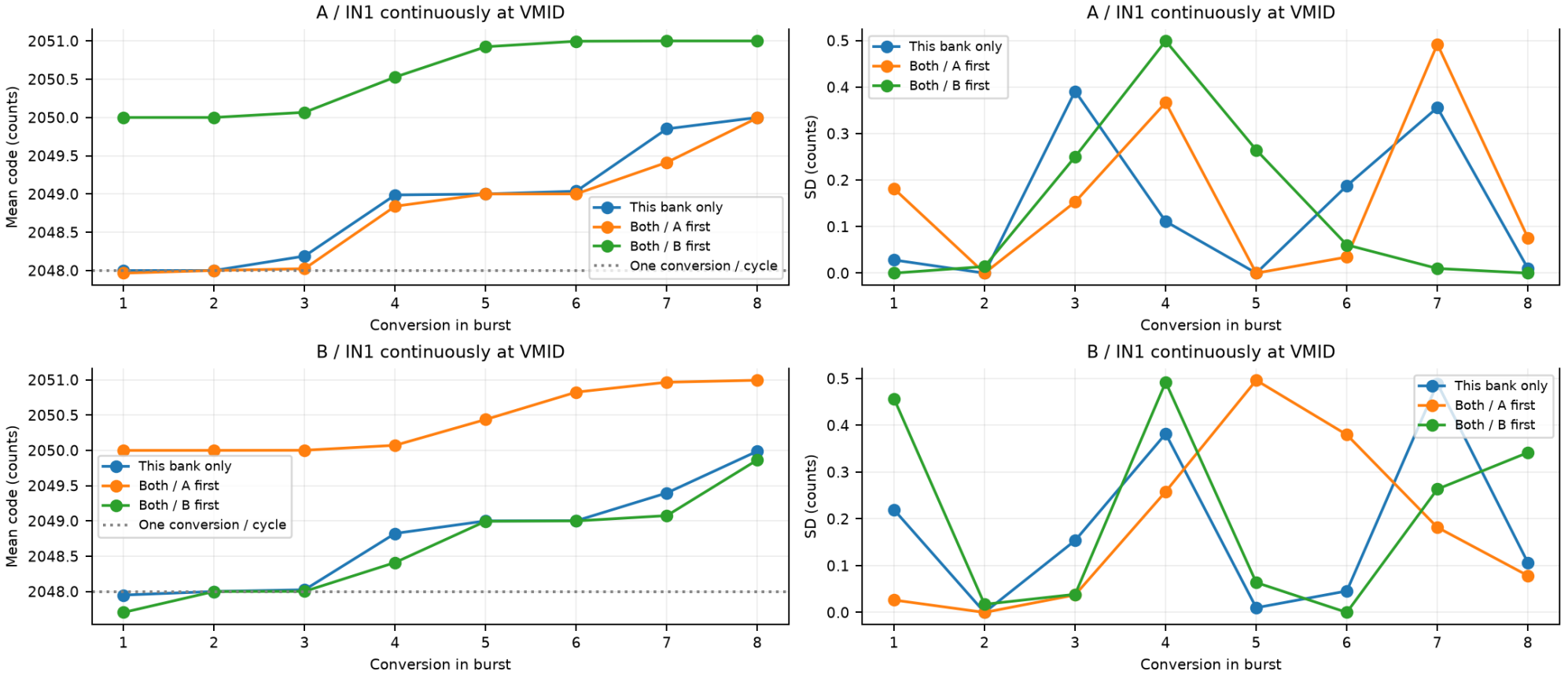
The preconditioning interval is fixed at approximately 100 ms, without a recorded voltage immediately before disconnection. Consequently, compare these as repeatable reconnection responses, not direct measurements of isolated sensor voltage, a guaranteed fully charged initial state, or an identified leakage resistance/capacitance. MUX off-state leakage, PCB leakage and the sensor can all contribute.

Known-voltage control: address 6 and 7 are the same VMID net

Delay us	Array/MUX	Held address 6	Held address 7	Toggle: address 6	Toggle: address 7	Toggle SD, addr6 / 7
1	A/1	2,048.00	2,048.00	2,048.00	2,048.00	0.000 / 0.017
1	A/2	2,048.00	2,048.00	2,048.00	2,048.00	0.095 / 0.045
1	A/3	2,049.00	2,049.00	2,049.00	2,049.00	0.383 / 0.481
1	B/1	2,049.00	2,049.00	2,049.00	2,049.00	0.024 / 0.073
1	B/2	2,049.00	2,049.00	2,049.00	2,049.00	0.034 / 0.000
1	B/3	2,050.00	2,050.00	2,049.00	2,049.00	0.487 / 0.344
5	A/1	2,048.00	2,048.00	2,048.00	2,048.00	0.000 / 0.000
5	A/2	2,048.00	2,048.00	2,048.00	2,048.00	0.135 / 0.067
5	A/3	2,049.00	2,049.00	2,049.00	2,049.00	0.345 / 0.452
5	B/1	2,049.00	2,049.00	2,049.00	2,049.00	0.000 / 0.053
5	B/2	2,049.00	2,049.00	2,049.00	2,049.00	0.041 / 0.000
5	B/3	2,050.00	2,050.00	2,049.00	2,049.00	0.497 / 0.376
50	A/1	2,048.00	2,048.00	2,048.00	2,048.00	0.000 / 0.000
50	A/2	2,048.00	2,048.00	2,048.00	2,048.00	0.156 / 0.075
50	A/3	2,049.00	2,049.00	2,049.00	2,049.00	0.312 / 0.446
50	B/1	2,049.00	2,049.00	2,049.00	2,049.00	0.000 / 0.067
50	B/2	2,049.00	2,049.00	2,049.00	2,049.00	0.029 / 0.000
50	B/3	2,050.00	2,050.00	2,049.00	2,049.00	0.497 / 0.357

Only ADC inputs 1-3 are read; each has address 6 physically tied to its address-7 VMID network. MUX4 still switches sensor 6 versus VMID because address/EN lines are shared, so this is a whole-board low-impedance control rather than a pure isolated GPIO test. Requested cycle period 710 us; held case reads once/cycle, toggled case twice. Nominal VMID is 2048 counts.

Reference-only bursts: no external switching or sensor transition



External address remains 7 throughout acquisition and ADC input remains IN1. No external MUX or ADC-input switching occurs after preparation. Fixed 120 us reference-wait phase plus 100 us read slot, period 250 us. Means expose sub-count code occupancy; values are medians of three per-record means. A within-burst shift here cannot be attributed to a sensor transition. It does not uniquely distinguish ADC behavior, buffer drive, shared VREF/VBIAS transients or digital coupling.

Shared conversion history and the VBIAS / VREF ratio

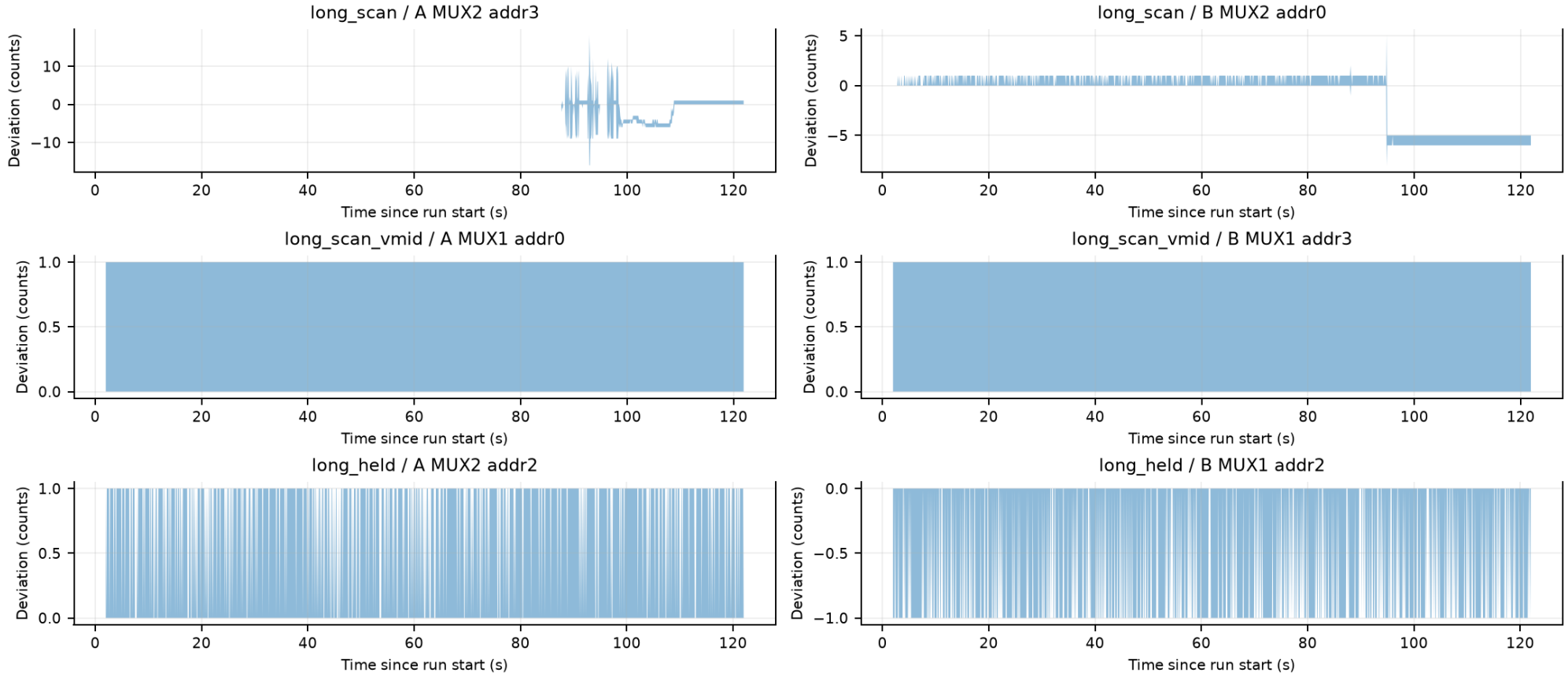
Bank activity	Array	First mean code	Eighth mean code	Change counts	Single-bank one-conversion mean
A only	A	2,047.9992	2,049.9999	2.0007	2,048.0000
B only	B	2,047.9493	2,049.9886	2.0393	2,047.9996
A then B	A	2,047.9659	2,049.9942	2.0283	2,048.0000
A then B	B	2,049.9993	2,050.9938	0.9945	2,047.9996
B then A	B	2,047.7033	2,049.8651	2.1618	2,047.9996
B then A	A	2,050.0000	2,051.0000	1.0000	2,048.0000

All inputs in this control are held on a low-impedance reference connection. A change that follows conversion position, including which ADC bank was read first, is evidence of shared conversion history. It cannot be assigned solely to external MUX switching or PZT charging. ADC behavior, buffer output drive, reference/supply transients and digital coupling remain possible contributors.

Near midscale, N approximately equals 4096 x VBIAS / VREF. Small changes obey: delta-N approximately equals (4096 / 3.3) x (delta-VBIAS - 0.5 x delta-VREF). For illustration, a +3-count change could correspond to VBIAS/buffer output rising about 2.42 mV with VREF fixed, or VREF falling about 4.83 mV with VBIAS fixed. These are equivalent-voltage calculations, not rail measurements.

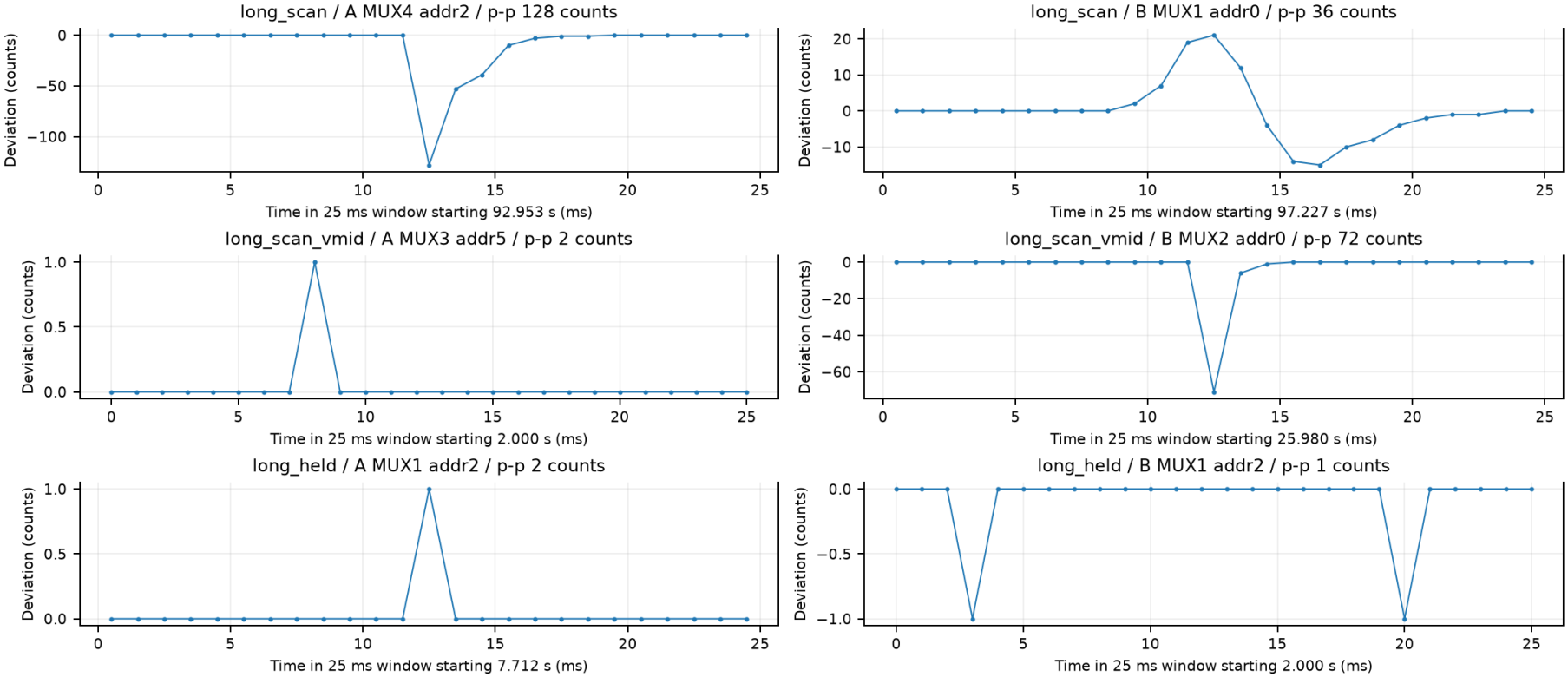
The ADC124S101 uses its analog supply VA as its reference and draws conversion-dependent current. REF2033 VREF and VBIAS sharing an internal source does not guarantee cancellation of different fast output-load disturbances. Simultaneous probing of ADC VA/VREF, VMID and buffer output would distinguish these possibilities. The report does not identify a defective component or demonstrate a production fix.

Long records: time traces and spikes



The highest-SD route in each array/condition is shown. Each 100 ms envelope retains minimum and maximum samples, so spikes are not hidden by averaging. Warm-up is excluded. These are 120-second recordings.

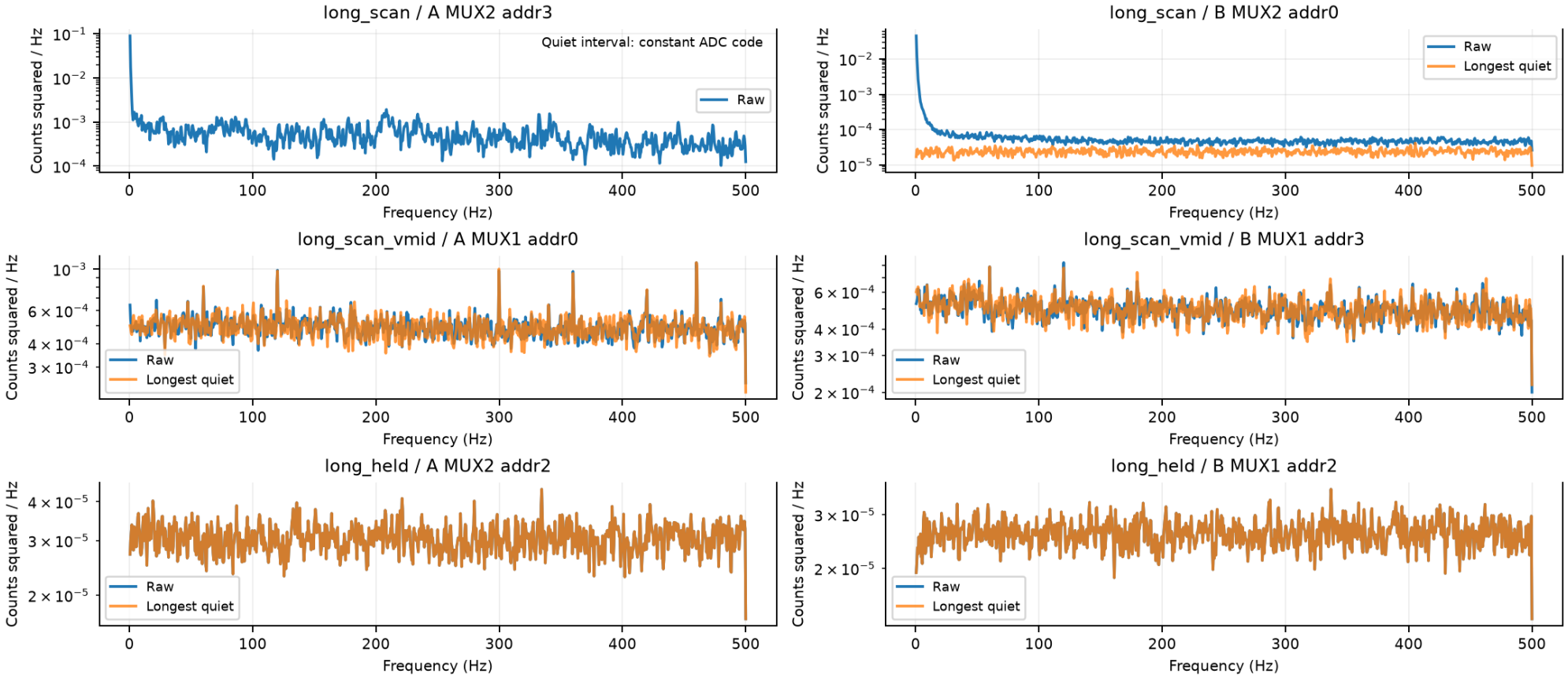
Largest excursions: independently selected by peak-to-peak range



Condition	Array/MUX/address	Raw p-p counts / mV	Raw SD counts / mV
long_scan	A/4/2	128 / 103.125	0.4387 / 0.3535
long_scan	B/1/0	36 / 29.004	0.1538 / 0.1239
long_scan_vmid	A/3/5	2 / 1.611	0.0835 / 0.0673
long_scan_vmid	B/2/0	72 / 58.008	0.2059 / 0.1659
long_held	A/1/2	2 / 1.611	0.0104 / 0.0084
long_held	B/1/2	1 / 0.806	0.1144 / 0.0922

Selection maximizes the full-record peak-to-peak range independently of the SD-based selections. Every sampled point in the 25 ms window is shown. A one-count excursion may be ordinary ADC code toggling; a large range alone does not establish whether an event was electrical noise or a real PZT response.

Long records: power spectral density



Welch PSD, up to 2048 samples/window, constant detrending. Only contiguous received sequence blocks are used; USB omissions are not joined or interpolated. Constant-code quiet intervals have zero recorded PSD and are annotated; they do not establish a zero analog-noise floor. Peak locations alone do not identify the interference source.

Long-record noise and shared correlations

Condition	Array/MUX/address	Raw SD counts / mV	Raw p-p counts / mV	Quiet SD counts / mV	Quiet p-p counts	Quiet seconds
long_scan	A/2/3	1.4227 / 1.1462	34 / 27.393	0.0000 / 0.0000	0	43.00
long_scan	B/2/0	2.1273 / 1.7139	13 / 10.474	0.1084 / 0.0873	1	43.00
long_scan_vmid	A/1/0	0.4993 / 0.4023	1 / 0.806	0.4996 / 0.4025	1	95.00
long_scan_vmid	B/1/3	0.4999 / 0.4028	1 / 0.806	0.5000 / 0.4028	1	95.00
long_held	A/2/2	0.1242 / 0.1001	1 / 0.806	0.1242 / 0.1001	1	120.00
long_held	B/1/2	0.1144 / 0.0922	1 / 0.806	0.1144 / 0.0922	1	120.00

Common-quiet statistics use the prior report's separated-state detector: a dominant one-code (60%) or adjacent two-code (80%) state anchors each route; allow a one-count margin. A separated excursion needs at least 1% tail mass and a peak at least 0.5% of samples and three times the edge density. At least two routes must depart in the same sweep; that sweep and one neighbor each side are rejected identically for every route. A sustained-burst screen also rejects any one-second bin whose SD on any route exceeds max(0.75 count, four times that route's median bin SD), padding one bin each side. Then select the longest remaining uninterrupted received interval; do not join different baseline states. This is duration selection, not selection for minimum SD. Isolated spikes inside that interval remain. Raw traces and raw statistics retain every event. Exact intervals and all routes' quiet statistics are in long_quiet_channels.csv.

Condition	Route 1	Route 2	Raw correlation	Within-second correlation
long_scan	2:4:5	2:1:5	-0.515	-0.511
long_scan	1:3:2	1:2:2	0.477	0.476
long_scan	1:4:2	1:2:2	0.329	0.331
long_scan	2:4:1	2:2:1	0.244	0.246
long_scan	1:4:2	1:3:2	0.243	0.245
long_scan_vmid	1:3:0	1:1:0	0.233	0.223
long_scan_vmid	1:4:0	1:1:0	0.230	0.219
long_scan_vmid	1:4:0	1:3:0	0.218	0.205
long_scan_vmid	1:2:0	1:1:0	0.211	0.202
long_scan	1:3:5	1:1:5	-0.375	-0.168
long_scan	2:3:2	2:1:2	-0.145	-0.145
long_scan_vmid	1:3:2	1:1:2	0.109	0.109
long_scan	1:3:5	1:2:5	0.081	0.089
long_scan_vmid	2:4:1	2:3:1	0.089	0.087

Pairs are restricted to the same external-address phase; inputs are acquired sequentially, not perfectly simultaneously. Only routes with SD above 0.05 count are compared. Within-second correlation subtracts each route's mean in each one-second bin. These are descriptive associations, with multiple comparisons; they do not establish a physical coupling source.

The unreset scan record contains a burst/step event late in the record. The later VMID and held-address records are quieter, but one record per condition at different times cannot establish whether mode or an uninstrumented external disturbance caused that difference.

Capture integrity and firmware restoration

Check	Result
Expected / complete captures	762 / 762
Acquired / received / discarded cycles	7,919,928 / 7,919,811 / 117
Transfer / data / USB write errors	[0, 0, 0]
Controlled timing overruns	0
Frame integrity	All captured frames decoded with ADG1 schema, physical event labels, monotonic sequence and FNV-1a checksum
Frozen production image	Original HEX and settings backed up before diagnostic upload; restore uses that frozen HEX
Restored firmware / stopped	adc124-selector-lookahead-v1 / false
Restored engine / scan order	blocking / adc
Restoration state	production_restored_verified

The diagnostic environment excludes production main.cpp. It retains the production GPIO switching guards and LPSPI driver, and adds arbitrary external-address programs, input/bank ordering, isolated waits and retained repeated conversions. Its identity and binary magic differ from production; desktop GUI parsing is unchanged.

Validation: 810 tagged diagnostic configurations with three consecutive-cycle checks each; malformed commands, isolation, checksum and failure checks. Production host-model regression also passed. All physical channel labels in quiet recordings still rely on the documented board wiring; no external stimulus was applied to establish analog channel identity.

Long record / array	MUX/address	Delay median us	Delay range ns	Burst duration range ns	Code/delay correlation	Code/duration correlation
long_scan / A	2/3	5.0767	5.00	5.00	0.0009	-0.0051
long_scan / B	2/0	10.3433	10.00	1.67	-0.0024	-0.0015
long_scan_vmid / A	1/0	5.0767	5.00	13.33	0.0032	0.0044
long_scan_vmid / B	1/3	10.3417	11.67	10.00	0.0051	-0.0012
long_held / A	2/2	5.0750	3.33	6.67	-0.0010	0.0010
long_held / B	1/2	10.3433	6.67	6.67	-0.0003	-0.0020

These compare the SD-selected routes against logged per-bank burst timing. A constant marker has undefined correlation (dash). Markers do not resolve actual pin edges or sample apertures; absence of correlation does not rule out electrical USB/supply coupling.

Device references and limits of causal attribution

ADC124S101 data sheet - https://www.ti.com/lit/ds/symlink/adc124s101.pdf

TMUX1108 data sheet - https://www.ti.com/lit/ds/symlink/tmux1108.pdf

TMUX1308A data sheet - https://www.ti.com/lit/ds/symlink/tmux1308a.pdf

OPA365 data sheet - https://www.ti.com/lit/ds/symlink/opa365.pdf

REF2033 family data sheet - https://www.ti.com/lit/pdf/sbos600

TI high-impedance MUX settling application note - https://www.ti.com/lit/pdf/SCDA054

ADC124S101 uses a three-clock acquisition interval and a typical 30 pF sampling capacitor; its data sheet explicitly does not require dummy conversions for power-up. More conversions changing a measured board baseline therefore does not establish a digital protocol requirement to discard a sample. They also change sampling-load activity and supply-current history. The OPA365 separates the external MUX node from the ADC sampling capacitor; directly multiplying that capacitor by the external bias resistor is inappropriate.

These software-controlled experiments do not separately measure MUX injection charge, PZT capacitance, pin leakage, reference ripple, buffer stability or physical GPIO/SPI waveforms. MUX type is tied to array location and associated components. The retained data support comparisons between operating histories; component-level attribution requires a controlled hardware substitution or electrical probing.

Program manifest and timing

Program	Family	Reps	Clock MHz	Banks	Steps	Requested period us	Measured median period us
boundary_address6_1	boundary_matched	3	16	3	7	710	710.00
boundary_address6_50	boundary_matched	3	16	3	7	710	710.00
boundary_address6_5	boundary_matched	3	16	3	7	710	710.00
boundary_disabled_1	boundary_matched	3	16	3	7	710	710.00
boundary_disabled_50	boundary_matched	3	16	3	7	710	710.00
boundary_disabled_5	boundary_matched	3	16	3	7	710	710.00
boundary_hold_1	boundary_matched	3	16	3	7	710	710.00
boundary_hold_50	boundary_matched	3	16	3	7	710	710.00
boundary_hold_5	boundary_matched	3	16	3	7	710	710.00
boundary_vmid_1	boundary_matched	3	16	3	7	710	710.00
boundary_vmid_50	boundary_matched	3	16	3	7	710	710.00
boundary_vmid_5	boundary_matched	3	16	3	7	710	710.00
boundary_vmid_adc_1	boundary_matched	3	16	3	7	710	710.00
boundary_vmid_adc_50	boundary_matched	3	16	3	7	710	710.00
boundary_vmid_adc_5	boundary_matched	3	16	3	7	710	710.00
fast_core_1	boundary_fast	3	16	3	6	0	78.00
fast_core_50	boundary_fast	3	16	3	6	0	372.00
fast_core_5	boundary_fast	3	16	3	6	0	102.00
fast_full_1	boundary_fast	3	16	3	7	0	85.00
fast_full_50	boundary_fast	3	16	3	7	0	428.00
fast_full_5	boundary_fast	3	16	3	7	0	113.00
fast_full_vmid_1	boundary_fast	3	16	3	14	0	171.00
fast_full_vmid_50	boundary_fast	3	16	3	14	0	856.00
fast_full_vmid_5	boundary_fast	3	16	3	14	0	226.00
fast_order_core_gray	fast_order	3	16	3	6	0	102.00
fast_order_full_descending	fast_order	3	16	3	7	0	113.00
fast_order_full_rotated	fast_order	3	16	3	7	0	113.00
held_a0_mask15	held	3	16	3	1	710	710.00
held_a0_mask1	held	3	16	3	1	710	710.00
held_a0_mask2	held	3	16	3	1	710	710.00
held_a0_mask4	held	3	16	3	1	710	710.00

Program	Family	Reps	Clock MHz	Banks	Steps	Requested period us	Measured median period us
held_a0_mask8	held	3	16	3	1	710	710.00
held_a2_mask15	held	3	16	3	1	710	710.00
held_a2_mask1	held	3	16	3	1	710	710.00
held_a2_mask2	held	3	16	3	1	710	710.00
held_a2_mask4	held	3	16	3	1	710	710.00
held_a2_mask8	held	3	16	3	1	710	710.00
held_a4_mask15	held_extra	3	16	3	1	710	710.00
held_a5_mask15	held	3	16	3	1	710	710.00
held_a5_mask1	held	3	16	3	1	710	710.00
held_a5_mask2	held	3	16	3	1	710	710.00
held_a5_mask4	held	3	16	3	1	710	710.00
held_a5_mask8	held	3	16	3	1	710	710.00
known_held6_1_banks3	known_vmid	3	16	3	1	710	710.00
known_held6_50_banks3	known_vmid	3	16	3	1	710	710.00
known_held6_5_banks3	known_vmid	3	16	3	1	710	710.00
known_held7_1_banks3	known_vmid	3	16	3	1	710	710.00
known_held7_50_banks3	known_vmid	3	16	3	1	710	710.00
known_held7_5_banks3	known_vmid	3	16	3	1	710	710.00
known_toggle67_1_banks3	known_vmid	3	16	3	2	710	710.00
known_toggle67_50_banks3	known_vmid	3	16	3	2	710	710.00
known_toggle67_5_banks1	known_vmid	3	16	1	2	710	710.00
known_toggle67_5_banks2	known_vmid	3	16	2	2	710	710.00
known_toggle67_5_banks3	known_vmid	3	16	3	2	710	710.00
leak_b0_m0_1000000	leakage	3	16	1	3	1100500	1,100,500.00
leak_b0_m0_100000	leakage	3	16	1	3	200500	200,500.00
leak_b0_m0_10000	leakage	3	16	1	3	110500	110,500.00
leak_b0_m0_1000	leakage	3	16	1	3	101500	101,500.00
leak_b0_m1_1000000	leakage	3	16	1	3	1100500	1,100,500.00
leak_b0_m1_100000	leakage	3	16	1	3	200500	200,500.00
leak_b0_m1_10000	leakage	3	16	1	3	110500	110,500.00
leak_b0_m1_1000	leakage	3	16	1	3	101500	101,500.00
leak_b0_m2_1000000	leakage	3	16	1	3	1100500	1,100,500.00
leak_b0_m2_100000	leakage	3	16	1	3	200500	200,500.00

Program	Family	Reps	Clock MHz	Banks	Steps	Requested period us	Measured median period us
leak_b0_m2_10000	leakage	3	16	1	3	110500	110,500.00
leak_b0_m2_1000	leakage	3	16	1	3	101500	101,500.00
leak_b0_m3_1000000	leakage	3	16	1	3	1100500	1,100,500.00
leak_b0_m3_100000	leakage	3	16	1	3	200500	200,500.00
leak_b0_m3_10000	leakage	3	16	1	3	110500	110,500.00
leak_b0_m3_1000	leakage	3	16	1	3	101500	101,500.00
leak_b1_m0_1000000	leakage	3	16	2	3	1100500	1,100,500.00
leak_b1_m0_100000	leakage	3	16	2	3	200500	200,500.00
leak_b1_m0_10000	leakage	3	16	2	3	110500	110,500.00
leak_b1_m0_1000	leakage	3	16	2	3	101500	101,500.00
leak_b1_m1_1000000	leakage	3	16	2	3	1100500	1,100,500.00
leak_b1_m1_100000	leakage	3	16	2	3	200500	200,500.00
leak_b1_m1_10000	leakage	3	16	2	3	110500	110,500.00
leak_b1_m1_1000	leakage	3	16	2	3	101500	101,500.00
leak_b1_m2_1000000	leakage	3	16	2	3	1100500	1,100,500.00
leak_b1_m2_100000	leakage	3	16	2	3	200500	200,500.00
leak_b1_m2_10000	leakage	3	16	2	3	110500	110,500.00
leak_b1_m2_1000	leakage	3	16	2	3	101500	101,500.00
leak_b1_m3_1000000	leakage	3	16	2	3	1100500	1,100,500.00
leak_b1_m3_100000	leakage	3	16	2	3	200500	200,500.00
leak_b1_m3_10000	leakage	3	16	2	3	110500	110,500.00
leak_b1_m3_1000	leakage	3	16	2	3	101500	101,500.00
long_held	long_noise	1	16	3	1	1000	1,000.00
long_scan	long_noise	1	16	3	6	1000	1,000.00
long_scan_vmid	long_noise	1	16	3	12	1000	1,000.00
order_ascending_ref0	order	3	16	3	7	710	710.00
order_ascending_ref1	order	3	16	3	7	710	710.00
order_descending_ref0	order	3	16	3	7	710	710.00
order_descending_ref1	order	3	16	3	7	710	710.00
order_gray_ref0	order	3	16	3	7	710	710.00
order_gray_ref1	order	3	16	3	7	710	710.00
order_permuted_ref0	order	3	16	3	7	710	710.00
order_permuted_ref1	order	3	16	3	7	710	710.00

Program	Family	Reps	Clock MHz	Banks	Steps	Requested period us	Measured median period us
order_rotated_ref0	order	3	16	3	7	710	710.00
order_rotated_ref1	order	3	16	3	7	710	710.00
readorder_1_banks1_first0_order228	readorder	3	16	1	7	710	710.00
readorder_1_banks2_first0_order228	readorder	3	16	2	7	710	710.00
readorder_1_banks3_first0_order228	readorder	3	16	3	7	710	710.00
readorder_1_banks3_first0_order27	readorder	3	16	3	7	710	710.00
readorder_1_banks3_first1_order228	readorder	3	16	3	7	710	710.00
readorder_50_banks1_first0_order228	readorder	3	16	1	7	710	710.00
readorder_50_banks2_first0_order228	readorder	3	16	2	7	710	710.00
readorder_50_banks3_first0_order228	readorder	3	16	3	7	710	710.00
readorder_50_banks3_first0_order27	readorder	3	16	3	7	710	710.00
readorder_50_banks3_first1_order228	readorder	3	16	3	7	710	710.00
readorder_5_banks1_first0_order228	readorder	3	16	1	7	710	710.00
readorder_5_banks2_first0_order228	readorder	3	16	2	7	710	710.00
readorder_5_banks3_first0_order228	readorder	3	16	3	7	710	710.00
readorder_5_banks3_first0_order27	readorder	3	16	3	7	710	710.00
readorder_5_banks3_first1_order228	readorder	3	16	3	7	710	710.00
recovery_b0_m0_16000000_1_eight	recovery	3	16	1	2	250	250.00
recovery_b0_m0_16000000_1_wait_plus	recovery_bracket	3	16	1	2	250	250.00
recovery_b0_m0_16000000_1_wait	recovery	3	16	1	2	250	250.00
recovery_b0_m0_16000000_50_eight	recovery	3	16	1	2	250	250.00
recovery_b0_m0_16000000_50_wait_plus	recovery_bracket	3	16	1	2	250	250.00
recovery_b0_m0_16000000_50_wait	recovery	3	16	1	2	250	250.00
recovery_b0_m0_16000000_5_eight	recovery	3	16	1	2	250	250.00
recovery_b0_m0_16000000_5_wait_plus	recovery_bracket	3	16	1	2	250	250.00
recovery_b0_m0_16000000_5_wait	recovery	3	16	1	2	250	250.00
recovery_b0_m0_8000000_1_eight	recovery	3	8	1	2	250	250.00
recovery_b0_m0_8000000_1_wait_plus	recovery_bracket	3	8	1	2	250	250.00
recovery_b0_m0_8000000_1_wait	recovery	3	8	1	2	250	250.00
recovery_b0_m0_8000000_5_eight	recovery	3	8	1	2	250	250.00
recovery_b0_m0_8000000_5_wait_plus	recovery_bracket	3	8	1	2	250	250.00
recovery_b0_m0_8000000_5_wait	recovery	3	8	1	2	250	250.00
recovery_b0_m1_16000000_1_eight	recovery	3	16	1	2	250	250.00

Program	Family	Reps	Clock MHz	Banks	Steps	Requested period us	Measured median period us
recovery_b0_m1_16000000_1_wait_plus	recovery_bracket	3	16	1	2	250	250.00
recovery_b0_m1_16000000_1_wait	recovery	3	16	1	2	250	250.00
recovery_b0_m1_16000000_50_eight	recovery	3	16	1	2	250	250.00
recovery_b0_m1_16000000_50_wait_plus	recovery_bracket	3	16	1	2	250	250.00
recovery_b0_m1_16000000_50_wait	recovery	3	16	1	2	250	250.00
recovery_b0_m1_16000000_5_eight	recovery	3	16	1	2	250	250.00
recovery_b0_m1_16000000_5_wait_plus	recovery_bracket	3	16	1	2	250	250.00
recovery_b0_m1_16000000_5_wait	recovery	3	16	1	2	250	250.00
recovery_b0_m1_8000000_1_eight	recovery	3	8	1	2	250	250.00
recovery_b0_m1_8000000_1_wait_plus	recovery_bracket	3	8	1	2	250	250.00
recovery_b0_m1_8000000_1_wait	recovery	3	8	1	2	250	250.00
recovery_b0_m1_8000000_5_eight	recovery	3	8	1	2	250	250.00
recovery_b0_m1_8000000_5_wait_plus	recovery_bracket	3	8	1	2	250	250.00
recovery_b0_m1_8000000_5_wait	recovery	3	8	1	2	250	250.00
recovery_b0_m2_16000000_1_eight	recovery	3	16	1	2	250	250.00
recovery_b0_m2_16000000_1_wait_plus	recovery_bracket	3	16	1	2	250	250.00
recovery_b0_m2_16000000_1_wait	recovery	3	16	1	2	250	250.00
recovery_b0_m2_16000000_50_eight	recovery	3	16	1	2	250	250.00
recovery_b0_m2_16000000_50_wait_plus	recovery_bracket	3	16	1	2	250	250.00
recovery_b0_m2_16000000_50_wait	recovery	3	16	1	2	250	250.00
recovery_b0_m2_16000000_5_eight	recovery	3	16	1	2	250	250.00
recovery_b0_m2_16000000_5_wait_plus	recovery_bracket	3	16	1	2	250	250.00
recovery_b0_m2_16000000_5_wait	recovery	3	16	1	2	250	250.00
recovery_b0_m2_8000000_1_eight	recovery	3	8	1	2	250	250.00
recovery_b0_m2_8000000_1_wait_plus	recovery_bracket	3	8	1	2	250	250.00
recovery_b0_m2_8000000_1_wait	recovery	3	8	1	2	250	250.00
recovery_b0_m2_8000000_5_eight	recovery	3	8	1	2	250	250.00
recovery_b0_m2_8000000_5_wait_plus	recovery_bracket	3	8	1	2	250	250.00
recovery_b0_m2_8000000_5_wait	recovery	3	8	1	2	250	250.00
recovery_b0_m3_16000000_1_eight	recovery	3	16	1	2	250	250.00
recovery_b0_m3_16000000_1_wait_plus	recovery_bracket	3	16	1	2	250	250.00
recovery_b0_m3_16000000_1_wait	recovery	3	16	1	2	250	250.00
recovery_b0_m3_16000000_50_eight	recovery	3	16	1	2	250	250.00

Program	Family	Reps	Clock MHz	Banks	Steps	Requested period us	Measured median period us
recovery_b0_m3_16000000_50_wait_plus	recovery_bracket	3	16	1	2	250	250.00
recovery_b0_m3_16000000_50_wait	recovery	3	16	1	2	250	250.00
recovery_b0_m3_16000000_5_eight	recovery	3	16	1	2	250	250.00
recovery_b0_m3_16000000_5_wait_plus	recovery_bracket	3	16	1	2	250	250.00
recovery_b0_m3_16000000_5_wait	recovery	3	16	1	2	250	250.00
recovery_b0_m3_8000000_1_eight	recovery	3	8	1	2	250	250.00
recovery_b0_m3_8000000_1_wait_plus	recovery_bracket	3	8	1	2	250	250.00
recovery_b0_m3_8000000_1_wait	recovery	3	8	1	2	250	250.00
recovery_b0_m3_8000000_5_eight	recovery	3	8	1	2	250	250.00
recovery_b0_m3_8000000_5_wait_plus	recovery_bracket	3	8	1	2	250	250.00
recovery_b0_m3_8000000_5_wait	recovery	3	8	1	2	250	250.00
recovery_b1_m0_16000000_1_eight	recovery	3	16	2	2	250	250.00
recovery_b1_m0_16000000_1_wait_plus	recovery_bracket	3	16	2	2	250	250.00
recovery_b1_m0_16000000_1_wait	recovery	3	16	2	2	250	250.00
recovery_b1_m0_16000000_50_eight	recovery	3	16	2	2	250	250.00
recovery_b1_m0_16000000_50_wait_plus	recovery_bracket	3	16	2	2	250	250.00
recovery_b1_m0_16000000_50_wait	recovery	3	16	2	2	250	250.00
recovery_b1_m0_16000000_5_eight	recovery	3	16	2	2	250	250.00
recovery_b1_m0_16000000_5_wait_plus	recovery_bracket	3	16	2	2	250	250.00
recovery_b1_m0_16000000_5_wait	recovery	3	16	2	2	250	250.00
recovery_b1_m0_8000000_1_eight	recovery	3	8	2	2	250	250.00
recovery_b1_m0_8000000_1_wait_plus	recovery_bracket	3	8	2	2	250	250.00
recovery_b1_m0_8000000_1_wait	recovery	3	8	2	2	250	250.00
recovery_b1_m0_8000000_5_eight	recovery	3	8	2	2	250	250.00
recovery_b1_m0_8000000_5_wait_plus	recovery_bracket	3	8	2	2	250	250.00
recovery_b1_m0_8000000_5_wait	recovery	3	8	2	2	250	250.00
recovery_b1_m1_16000000_1_eight	recovery	3	16	2	2	250	250.00
recovery_b1_m1_16000000_1_wait_plus	recovery_bracket	3	16	2	2	250	250.00
recovery_b1_m1_16000000_1_wait	recovery	3	16	2	2	250	250.00
recovery_b1_m1_16000000_50_eight	recovery	3	16	2	2	250	250.00
recovery_b1_m1_16000000_50_wait_plus	recovery_bracket	3	16	2	2	250	250.00
recovery_b1_m1_16000000_50_wait	recovery	3	16	2	2	250	250.00
recovery_b1_m1_16000000_5_eight	recovery	3	16	2	2	250	250.00

Program	Family	Reps	Clock MHz	Banks	Steps	Requested period us	Measured median period us
recovery_b1_m1_16000000_5_wait_plus	recovery_bracket	3	16	2	2	250	250.00
recovery_b1_m1_16000000_5_wait	recovery	3	16	2	2	250	250.00
recovery_b1_m1_8000000_1_eight	recovery	3	8	2	2	250	250.00
recovery_b1_m1_8000000_1_wait_plus	recovery_bracket	3	8	2	2	250	250.00
recovery_b1_m1_8000000_1_wait	recovery	3	8	2	2	250	250.00
recovery_b1_m1_8000000_5_eight	recovery	3	8	2	2	250	250.00
recovery_b1_m1_8000000_5_wait_plus	recovery_bracket	3	8	2	2	250	250.00
recovery_b1_m1_8000000_5_wait	recovery	3	8	2	2	250	250.00
recovery_b1_m2_16000000_1_eight	recovery	3	16	2	2	250	250.00
recovery_b1_m2_16000000_1_wait_plus	recovery_bracket	3	16	2	2	250	250.00
recovery_b1_m2_16000000_1_wait	recovery	3	16	2	2	250	250.00
recovery_b1_m2_16000000_50_eight	recovery	3	16	2	2	250	250.00
recovery_b1_m2_16000000_50_wait_plus	recovery_bracket	3	16	2	2	250	250.00
recovery_b1_m2_16000000_50_wait	recovery	3	16	2	2	250	250.00
recovery_b1_m2_16000000_5_eight	recovery	3	16	2	2	250	250.00
recovery_b1_m2_16000000_5_wait_plus	recovery_bracket	3	16	2	2	250	250.00
recovery_b1_m2_16000000_5_wait	recovery	3	16	2	2	250	250.00
recovery_b1_m2_8000000_1_eight	recovery	3	8	2	2	250	250.00
recovery_b1_m2_8000000_1_wait_plus	recovery_bracket	3	8	2	2	250	250.00
recovery_b1_m2_8000000_1_wait	recovery	3	8	2	2	250	250.00
recovery_b1_m2_8000000_5_eight	recovery	3	8	2	2	250	250.00
recovery_b1_m2_8000000_5_wait_plus	recovery_bracket	3	8	2	2	250	250.00
recovery_b1_m2_8000000_5_wait	recovery	3	8	2	2	250	250.00
recovery_b1_m3_16000000_1_eight	recovery	3	16	2	2	250	250.00
recovery_b1_m3_16000000_1_wait_plus	recovery_bracket	3	16	2	2	250	250.00
recovery_b1_m3_16000000_1_wait	recovery	3	16	2	2	250	250.00
recovery_b1_m3_16000000_50_eight	recovery	3	16	2	2	250	250.00
recovery_b1_m3_16000000_50_wait_plus	recovery_bracket	3	16	2	2	250	250.00
recovery_b1_m3_16000000_50_wait	recovery	3	16	2	2	250	250.00
recovery_b1_m3_16000000_5_eight	recovery	3	16	2	2	250	250.00
recovery_b1_m3_16000000_5_wait_plus	recovery_bracket	3	16	2	2	250	250.00
recovery_b1_m3_16000000_5_wait	recovery	3	16	2	2	250	250.00
recovery_b1_m3_8000000_1_eight	recovery	3	8	2	2	250	250.00

Program	Family	Reps	Clock MHz	Banks	Steps	Requested period us	Measured median period us
recovery_b1_m3_8000000_1_wait_plus	recovery_bracket	3	8	2	2	250	250.00
recovery_b1_m3_8000000_1_wait	recovery	3	8	2	2	250	250.00
recovery_b1_m3_8000000_5_eight	recovery	3	8	2	2	250	250.00
recovery_b1_m3_8000000_5_wait_plus	recovery_bracket	3	8	2	2	250	250.00
recovery_b1_m3_8000000_5_wait	recovery	3	8	2	2	250	250.00
refburst_banks1_first0_n1	refburst	3	16	1	2	250	250.00
refburst_banks1_first0_n8	refburst	3	16	1	2	250	250.00
refburst_banks2_first0_n1	refburst	3	16	2	2	250	250.00
refburst_banks2_first0_n8	refburst	3	16	2	2	250	250.00
refburst_banks3_first0_n1	refburst	3	16	3	2	250	250.00
refburst_banks3_first0_n8	refburst	3	16	3	2	250	250.00
refburst_banks3_first1_n8	refburst	3	16	3	2	250	250.00
reset_control_idle_1	reset_control	3	16	3	12	920	920.00
reset_control_idle_50	reset_control	3	16	3	12	920	920.00
reset_control_idle_5	reset_control	3	16	3	12	920	920.00
reset_control_vmid_1	reset_control	3	16	3	12	920	920.00
reset_control_vmid_50	reset_control	3	16	3	12	920	920.00
reset_control_vmid_5	reset_control	3	16	3	12	920	920.00
reset_control_vmid_adc_1	reset_control	3	16	3	12	920	920.00
reset_control_vmid_adc_50	reset_control	3	16	3	12	920	920.00
reset_control_vmid_adc_5	reset_control	3	16	3	12	920	920.00
reset_short_idle_1	reset_short	3	16	3	12	300	300.00
reset_short_idle_5	reset_short	3	16	3	12	300	300.00
reset_short_vmid_1	reset_short	3	16	3	12	300	300.00
reset_short_vmid_5	reset_short	3	16	3	12	300	300.00
reset_short_vmid_adc_1	reset_short	3	16	3	12	300	300.00
reset_short_vmid_adc_5	reset_short	3	16	3	12	300	300.00